

SLNOM: Exploring the sound of mastication as a behavioral change strategy for rapid eating regulation

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ABSTRACT

Rapid eating is linked to numerous health problems, such as obesity and gastritis. In this study, we explore the possibility of using mastication sound as a novel behavior change strategy to subtly regulate rapid eating behavior. In particular, we present SLNOM, a system that can automatically detect chewing behavior using a convolutional neural network (CNN) model, and slow down the playback speed of real-time mastication sounds to implicitly modify eating behavior. Two empirical studies have been conducted to determine: 1) the threshold of sound volume and speed without user perception; and 2) the feasibility and effectiveness of SLNOM in changing eating behavior using a Wizard of Oz study. The result indicated that manipulation of chewing sound could modulate eating rate, bite size without cognitive and behavioral effort. We discussed how cognitive science could explain these findings and suggested how future eating interventions can be designed to take advantage of current exploration.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **User studies**; • **Applied computing** → **Health care information systems**.

KEYWORDS

behavior change, rapid eating, mastication sound, cognitive science

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1 INTRODUCTION

Obesity has become one of the most significant public health concerns which has been linked to a number of adverse health consequences such as diabetes mellitus, fatty liver disease, and poor mental health. Accumulating evidence shows positive associations

between eating rate (i.e., the speed of food consumption) and weight status [28, 33] across various age groups [12, 20] (e.g., from children to older adults), demographics [22, 40] (e.g., gender) and cultural backgrounds [4, 20, 21] (e.g., European, Asian, and American samples). The metabolic effects of slow eating can explain the mechanism underlying this association, i.e., eating slowly could lead to an increased glucose response, greater satiation (earlier meal termination), and longer meal memories. It is suggested that modifying eating rates might be a behavioral strategy for weight loss, without dietary restrictions or extra energy expenditure.

Eating rate is difficult to change as it is highly habitual [29]. Several approaches have been developed to slow down the eating rate to promote healthy eating. Multi-sensory prompts (e.g., visual, auditory, and vibrational) have been used most widely in HCI. For instance, *Eating2Pic* [26] which is a painting system that nudges against rapid eating behavior by coloring landscape pictures. *10s Fork* [16], is a smart fork that vibrates whenever an interval between two mouthfuls of less than 10s is detected. However, the limitation of current just-in-time interventions is that they heavily rely on users' conscious scrutiny and self-motivation to make a change [1, 17]. This reflective behavior change is sensitive to both internal and external factors, such as a person's mood, which could degrade the effectiveness of the intervention. Therefore, our primary research concern is how to develop a rapid eating intervention that is effective in facilitating eating behavior change without cognitive and behavioral efforts.

The effects of auditory properties such as tempo on temporal behavior have been widely studied. As an example, people unconsciously align their running and exercising pace with the music beats [9]. Compared to slow tempo music, fast music can decrease drinking time [24]. As a result of the synergistic relationship between body rhythms and specific auditory features, we envision that the same principle can be applied to the eating practice. That is, the chewing sound, one of the most prevalent sounds during natural eating, could serve as a specific auditory style to subtly alter eating behavior without conscious awareness and efforts.

In this study, we explored how subtle changes in the speed of playback of chewing sounds could facilitate eating behavior change and subsequently reduce eating rate. To be more specific, we first conducted an empirical study to investigate the threshold of volume and playback speed of a recorded mastication sound in an eating practice that could be too subtle to discernible. Based on the identified volume and playback speed, we proposed SLNOM, a slow nom (i.e., chewing sound) system that uses live mastication sounds as a means to promote behavioral change. Specifically, This system consists of a wearable detection device that detects, records, and

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processes real-life chewing sound at a pre-determined playback speed and volume, as well as an earphone that delivers a modified real-time chewing sound to the user in real-time, as shown in Figure 1. In order to investigate the feasibility of SLNOM, we conducted a Wizard of Oz (WoZ) study [8], which compared eating microstructure and user perception under two conditions: control and experiment without the restriction of technical implementation. Based on our empirical findings, our proposed system can (1) decrease bite size and eating rate without compromising appetite rating; (2) maintain the overall eating experience, compared to a normal eating setting. Our main contribution in this study includes (1) introducing the possibility of incorporating chewing sounds, a previously underexplored auditory stimulus, to facilitate eating behavior change; (2) interpreting how behavior change is achieved based on cognitive theories, and providing design recommendations that will allow practitioners and designers to develop more engaging and feasible eating behavior in a systematic manner.

2 RELATED WORK

2.1 Dual-process theory: automatic mind and reflective mind

Dual-process theories suggest that decision-making is the result of two mental processes: the automatic and the reflective [19]. The automatic is characterized as being fast, effortless, intuitive, and emotional. In this mode, individuals tend to make decisions rapidly and without conscious thought. While in the reflective one, decision-making is usually a slow, thoughtful, thorough, and knowledge-based process. Although both modes are constantly active and interdependent, it has been identified that the majority of our daily decisions are driven by automatic reasoning [31] and reflective reasoning is only invoked when the automatic system is inadequate. The lack of motivation and ability has been critiqued as the most important reasons why individuals fail to perform the behavior [14]. To alleviate these problems, researchers have developed a variety of theories and techniques to achieve behavior modification in an unaware manner. For instance, Lee et al. [23] leveraged the decoy effect to promote healthy choices on a snack ordering website by adding the inferior alternative (i.e., small withered apple) to an original set. Adam et al. [1] employed the Delboeuf illusion to create a virtual plate that attempted to influence an individual's perception of the volume of food on the plate. In line with this theory, our primary goal is to explore the manipulation of chewing sounds as an eating behavioral change technique by triggering the automatic mind.

2.2 Multidisciplinary approaches for eating rate intervention

Multidisciplinary interventions have been implemented in various settings to target eating rate regulation for weight management. Researchers in the food industry predominantly focus on modifying the food shape and textures (i.e., increasing the chewiness [2] and hardness [6] of food) to encourage small changes in oral processing behaviors, and consequently decrease eating rate and ad libitum energy intake. In clinical settings, external devices such as *Mandometer* [15], *Smart fork* [16], *Bite Counter* [34] which can record

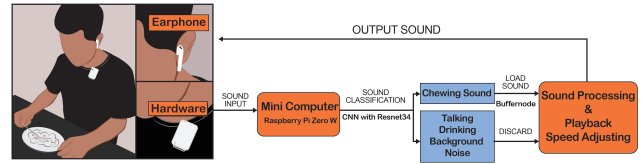


Figure 1: System diagram of SLNOM. A wearable device derives eating-related sound, a mini computer recognizes and modify chewing sound and deliver it to users via earphone.

people's oral processing behavior and provide visual or vibration prompts to guide individuals to modify their eating rate. Research in HCI focused on using non-obtrusive methods and modifying everyday objects in the eating environment (e.g., cutlery) to automatically detect eating behavior and provide real-time feedback. For instance, *Eating2Pic* [26] is a painting system that can reflect unhealthy eating behaviors such as rapid eating through the color of a digital canvas to encourage healthier eating. Zhang et al. [41] introduced a pneumatic fork that can change in rigidity by inflating and deflating the holder to prolong the time gap between mouthfuls. Noting that, these approaches have significant limitations. Food texture modification requires substantial dietary changes, which can be difficult if users do not want to change their diet. Using external prompters (i.e., visual or haptic), on the other hand, rely on users' perception and motivation for self-control, which may be tedious to maintain. In this regard, rather than direct instruction or enforcement, subtle modification to the eating context could be used to "nudge" eating rate change.

3 SYSTEM CONFIGURATION

Our previous exploration [7] designed a wearable device that used acoustic sensing to capture sound data and a convolutional neural network (CNN) to classify eating and non-eating episodes. As an extension of this technique, in this study, we mainly focus on how to integrate this designated device into SLNOM system to control chewing sound in real-time manner and explore its influence on eating behavior. To better understand the feasibility of this concept, we adopted the WoZ approach, which allows us to test design ideas without being constrained by the accuracy of the technical implementation. If the result of WoZ is favorable, we will then work on implementation to realize the concept.

4 EMPIRICAL STUDIES

To inform the system design, we considered two properties of the recorded chewing sounds: volume and playback speed. The initial idea is how the captured mastication sound could be processed so that people could perceive it as their own chewing sound during the eating experience.

4.1 Study 1(a): Threshold of volume without perception

Intuitively, there are discrepancies between sound perceived by individuals and sound transcribed by hardware. In this study, we aimed to determine the optimal volume of sound playback that could simulate the perceived chewing sound as closely as possible.

Participant and procedure We recruited 16 participants (eight females, aged between 21-27) using convenience sampling. They were asked to wear predesigned hardware in their collar to record their chewing sound. An earphone was used to broadcast the modified sound simultaneously. The delay between record and playback was less than 0.2s. The study was conducted in a private room that simulated a local restaurant with several tables and chairs. To mimic a normal lunch, participants were provided with a large portion of egg-fried rice, two fried vegetables (i.e., broccoli and cabbage), and a bowl of chicken broth as test food. The pilot study identified that recorded sound was always perceived as louder than real-time eating sound. Therefore, we set 100% as the maximum volume setting and gradually decreased the volume at the rate of 1% every 3 seconds. During the study, participants wore a wireless earphone in one ear in order to compare the sound they heard with the sound they perceived. They were required to raise their hands if they were unable to detect any difference.

Results The average volume of a recorded chewing sound that participants didn't recognize the difference was 72% (SD=6.9). To generalize the setting for all participants, instead of using the average result, we excluded the extreme data at both sides, and selected the minimum volume at 69% as the threshold.

4.2 Study 1(b): Threshold of speed change without perception

In this study, we investigated to what extent the playback speed of chewing sounds derived from a previously designed hardware device could be slowed, yet participants did not perceive any difference.

Participant and procedure Sixteen participants (seven females, age between 26 - 37) were recruited. Since the study aimed to regulate fast eaters' eating behavior, participants who reported eating rates below seven on a 10-point VAS scale were excluded. Similar to Study 1(a), eligible participants were invited into a private room and ate the provided food while wearing the predesigned device and an earphone throughout the entire study. During the study, we gradually slowed down the playback speed of chewing sounds collected from the hardware device at a rate of 1% per bite. A modified chewing sound was broadcast to participants via earphones. Participants were asked to raise their hands upon noticing any differences.

Results The perception of chewing speed varied from 0.79x to 0.92x. On average, participants were unable to discern a difference when the playback speed was decreased to 0.84x (SD = 0.037). It should be noted that our primary objective is to identify a threshold that can be applied to all participants. Therefore, after removing the extreme values at both ends, the maximum value (i.e., 0.87x) was selected as the threshold.

4.3 Study 2: Effectiveness of mastication sound manipulation on eating behavior

Based upon the findings, we further explored the effectiveness of playback speed of chewing sound (i.e., 0.87x speeds with 69% volume setting) on fast eaters' eating behavior. To be more specific, the experiment aimed to investigate three key research questions:

RQ1: To what extent could this system contribute to change in eating rate or other eating behavior?

RQ2: If the system could reduce the overall intake of food without affecting satiety levels?

RQ3: What are the users' perceptions of using SLNOM system during eating practice?

Experiment design This experiment was designed as a within-subject study, in which participants came to the lab for two visits: baseline and experiment during lunchtime (between 11:30 AM and 1:30 PM). We adopted the WoZ approach [8] to quickly test design ideas without technological restrictions. To clarify, instead of implementing the classifier model to recognize chewing, the chewing sounds were manually recorded and processed based on the predefined volume and speed in study1.

Participant 15 participants (10 males), aged between 26-33 with BMI (17.6 - 29.7) were recruited in this study. Exclusion criteria include medical conditions that would interfere with normal chewing and eating (e.g., anorexia nervosa) as well as allergies to the selected foods. During the recruitment process, participants were asked to rate their preference on whether they would like to eat all of them on a Likert scale of 10. Those who indicated a liking for food below 6 were also excluded.

Procedure To standardize satiety, participants were asked to refrain from eating, exercising, and drinking liquids other than water for at least 4 hours prior to participating in the study. Before and after each eating session, participants were asked to rate their satiety status on a VAS scale (details below). A large portion of test food (i.e., egg-fried rice, fried vegetables, and a bowl of chicken broth) accompanied by a plastic spoon was provided. In order to measure the real-time food consumption, food was served on a plate and placed on top of a concealed scale hidden under a table cloth. Each food was served in a larger portion, which is 50% larger than the standard portion size, in order to avoid "clear the plate" behavior [32]. Participants were instructed to eat until they were comfortably full. The entire experiment was video recorded for further analysis of eating behavior.

Measures *Oral processing behavior:* The oral processing behavior was analyzed using behavioral annotation software (ELAN, 6.1)[3]. Video-coding of oral processing behavior has been widely applied in the analysis of oral processing behavior without interrupting the real-life eating behavior [35]. The three key events of eating behavior were recorded (bite, chew, swallow) to obtain a set of oral processing parameters shown in Table 1. All oral processing behaviors were averaged across the meal.

Satiety rating: Before and after the eating session, participants were asked to report their perceived appetite in terms of hunger, fullness, and desire to eat with on a 100mm visual analog scales [11], anchored with a two-word description at 0 (not at all) and 100mm (extremely).

Results The assumption of normality was tested using Shapiro-Wilk ($p > .05$). Either paired-sample t-tests or Wilcoxon signed-rank tests (if normal distribution is violated) was performed on each parameter of oral processing behavior and satiety rating.

Oral processing behavior: There was a significant effect of chewing sound conditions on bite size, bite rate, eating rate, and active consumption, all at the $p < .05$ level. However, no difference was found in meal duration, chew rate, chews per bite, chews per gram,

Table 1: Annotation of eating behavior

Eating behavior	Definition
Meal duration (s)	Time between the first pick up spoon and the last drop of the spoon.
Oral exposure time (s)	Cumulative time food spent in the mouth between the bite and the swallow throughout the meal.
Oral exposure per bite (s/bite)	Total oral exposure time of food in mouth across the meal, divided by the total number of bites.
Eating rate (g/min)	Total amount of food consumed divided by the total oral exposure time.
Bite size (g/bite)	Total amount consumed divided by the total number of bites.
Bite rate (s/bite)	Total oral exposure time divided by the total number of bites.
Chews per bite	Total number of chews divided by total number of bites.
Chews per gram (chew/g)	Total number of chews divided by total amount consumed.
Chew rate (chew/s)	Number of chews divided by oral exposure time.
Average length of pause/ interval	Total non-active eating duration divided by total number of interval
Percentage of actively consuming food	The ratio of total oral exposure time to the total meal duration

the average length of pause/ interval between two conditions, see Figure 2. In particular, participants had smaller bite sizes when using the SLNOM system ($M=9.20$, $SD=1.69$) compared with baseline conditions ($M=8.08$, $SD=2.61$). Consequently, the eating rate was reduced in the SLNOM condition by 8.7% ($t(14)=89$, $p=0.046$), despite the bite rate in the baseline condition having increased by 9.6% ($t(14)=98$, $p=0.032$).

These results provide answers to **RQ1**, namely that SLNOM could have a significant impact on oral processing behaviors by decreasing the bite size, and decreasing the eating rate.

Food intake and Satiety status: There was no significant difference in the food intake between two experiment conditions, $t(14)=62$, $p=0.93$, although food intake decreased a bit ($M=309.93$, $SD=32.85$) at the SLNOM than baseline condition ($M=311.00$, $SD=31.179$). In addition, there was no significant difference of chewing sound conditions in satiety rating in terms of hunger ($t(14)=0$, $p=1.0$), fullness ($t(14)=-0.993$, $p=.34$), and desire to eat ($t(14)=2.08$, $p=.056$). These results provide initial evidence to answer **RQ2**, that SLNOM had no effect on the overall food intake and perceived satiety in a standardized meal.

User perception: For **RQ3**, on average, participants did not feel distracted when using SLNOM system and they reported similar satisfaction compared to the control condition. In the majority of cases (8/15), participants reported that the sound is negligible. As P1 stated, "I didn't notice any abnormality during the eating.". Some participants (5/15) failed to make the connection between the chewing sound and their own eating behavior, as a result, they didn't pay attention to the sound. As P2 explained, "I didn't recognize the sound as my chewing sound, therefore, I just left it in the background.". One participant stated that listening to chewing sounds makes her feel hungry and satisfied while eating. However, one participant commented that hearing the chewing noise could negatively affect her eating experience. As she explained, "When I listened to my own chewing sound, I may feel shame.". To alleviate any feeling of shamelessness or creepiness, she recommended the use of visual aids to illustrate what is going on in order to alleviate uncomfortable feelings.

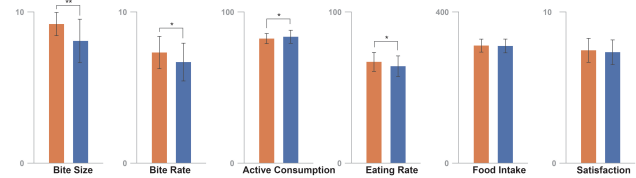


Figure 2: Different measures forming oral processing parameters and user satisfaction during two experiment conditions (orange: baseline, blue: intervention)

5 DISCUSSION

This study explored the effects of a slower live chewing sound on eating rate and other eating behavior. The result indicated that participants decreased their eating rate along with smaller bite sizes, faster bite rates, and more active eating periods while using SLNOM. These findings are in accordance with the existing studies that suggest specific parameters of oral processing behavior, such as large bite size [2, 12, 21] are linked to rapid eating behavior and food intake. However, our study found that the overall food intake was comparable between the two conditions, contrary to previous studies, in which a decrease in eating rate contributed to a decrease in food intake. These seemingly conflicting results could be explained by the fact that, although our proposed system decreased the eating rate, a longer meal duration (i.e., more active eating period) and faster chewing rate, which meant there was more opportunity to eat, may compensate for the overall increase in food intake under the SLNOM condition.

This exploratory study provided initial evidence that slowing down the playback speed of live chewing sounds may have an impact on eating behavior. Here we present several possible explanations for the observed relationship between modified chewing sounds and eating behavior, derived from cognitive science and automatic mind theories.

Synchronization of bodily rhythmic patterns to external auditory stimuli. Research consistently indicates that music tempo affects time-related activities such as drinking [24] and running

[9]. For example, athletes tend to choose specific music with a certain beat to accompany their running pace [5, 39]. Therefore, a slower eating rate may be associated with the fact that participants unconsciously align their chewing rhythm with the modified chewing rate. Interestingly, none of the participants found that the chewing sounds they heard were slowed down. This suggests that if the slowing down of the eating speed is caused by the beat of the chewing rate, it may be due to the auditory stimulus being too subtle to raise participants' awareness.

Cognitive incongruity induced the eating behavior change.

Cognition has been demonstrated to play a role in eating rate, such as eating while distracted media usage [37], poorer inhibitory control [13]. Specifically, some studies have focused on the effect of cognitive incongruity on eating habits. For instance, Mcleod et al. [25] found that participants tend to eat more slowly when food was served in incongruent conditions (i.e., pasta at breakfast) rather than the usual food-to-mealtime context (i.e., pasta at lunch). Our exploratory study suggests that the discrepancy between the self-perceived chewing sound and the transcribed chewing sound (mastication sound transmitted via earphone) may have created an interesting incongruity that resulted in the change in eating habits. As P3 stated, *"What I hear and what it records and plays may differ. It does not sound exactly like chewing."* Furthermore, due to the accumulation of the sound latency, the inconsistency between what the participants heard and the behavior they were performing became increasingly evident. P4 stated, *"It might not be real-time, because when I not chewing, the sound still rings."* In conclusion, this form of "unusual chewing sound" can be used as an effective strategy for altering behavior through the creation of a interesting eating environment to control eating in an unconscious manner.

5.1 Design recommendations

Integrating different auditory parameters to augment the eating sound. The purpose of this study was to subconsciously influence participants' eating pace by slowing the sound they heard themselves making while chewing. Few subjects, however, failed to identify the sound they heard as the sound of their own chewing. The reason for this is the low quality of live recordings. One option would be to record the high-quality chewing sounds in advance, then play them at a slower speed while detecting when people are eating. Additionally, studies have demonstrated that enhanced sound effects can enhance food texture and enhance the eating experience [10]. In light of these, we suggest that rather than using an individual's own chewing sound, the use of ASMRs eating sound [30] at an appropriate rate influences subjects' eating behavior and also help increase their feelings of satiety and meal enjoyment. Additionally, other auditory stimuli present in the eating scene, such as the sound of crashing dishes and the sound of whispers, can be used to develop a systematic understanding of their effects on eating behavior in future studies.

Incorporating underexplored factors and eating parameters for eating rate regulation. One systematic review concluded that eating rate is multidimensional and influenced by both individual [2, 6, 12, 21] and environmental levels [18, 29]. In this study, we only explored the influence of chewing rate as a potential factor contributing to a slower eating rate. Future studies are

encouraged to explore other associated factors (i.e., social and food environmental) in order to deliver a more engaging and effective intervention design. What's more, this study is concentrated on tackling rapid eating behavior which is highly associated with overweight and obesity. Note that, eating slowly is sometimes linked to underweight and malnutrition [27]. Intuitively, contrary to exploring strategies to slow down eating rate, further studies are warranted to investigate if the same principle, for instance, increased the playback speed of chewing sound could be implemented as a facilitator to induce a more energetic eating experience.

6 LIMITATION AND FUTURE WORK

In this study, we adopted the WoZ approach to quickly test and identify design parameters (i.e., volume and playback speed) of chewing sound. In the future, we intend to develop and implement the full-fledged system in real-world situations. Additionally, our previous research utilized an acoustic microphone to collect audio data that may not be applicable to real-world scenarios with background noise. Therefore, we will explore the combination of other modalities for eating detection, such as incorporating inertial sensing into wrist-worn devices for eating gesture detection [38] in the future to build a robust chewing detection system. What's more, eating is a multisensory experience [36]. In light of this, we envisioned the possibility of the incorporation with other modalities (e.g., visual) to facilitate eating behavior changes in a more intuitive and naturalistic way.

7 CONCLUSION

In this paper, we examined the effects of an auditory stimulus (the playback speed of chewing sounds) on rapid eating behavior. We conducted two empirical studies to test if the modified chewing sound can be used as a behavior change technique to help subtly change our eating behavior in an intuitive way. Results showed that the proposed concept was able to decrease the bite size and slow down the eating rate without affecting satiety and eating satisfaction. We discussed in detail how the chewing sounds contributed to eating behavioral changes through different theoretical explanations. Additionally, we provided design suggestions for future designers and practitioners to develop a more engaging and feasible eating rate intervention (i.e., either slowing down or speeding up) to promote healthy eating.

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